

Class Test: SOUND (Case Study Based)

Class: IX

Subject: Science

Time: 60 Minutes

Chapter: Sound

Maximum Marks: 40

General Instructions

1. All questions are **compulsory**.
2. The question paper consists of **20 case-study based questions**.
3. Each case study has **sub-parts** carrying **1–3 marks** based on difficulty.
4. Use of calculator is **not permitted**.
5. Draw neat, labelled diagrams wherever required.
6. Assume speed of sound in air = **340 m s^{-1}** unless stated otherwise.

Case Study 1 (1 Mark)

A student plucks a stretched rubber band and hears a sound.

Question: What is the basic cause of sound production?

Ans : _____

Case Study 2 (1 Mark)

An astronaut cannot hear any sound on the Moon.

Question: State the reason for this observation.

Ans : _____



Case Study 3 (2 Marks)

A bell is rung inside a glass jar from which air is gradually removed.

Questions:

- a) What happens to the sound as air is removed? (1)
- b) What does this prove about sound? (1)

Ans a) : _____

Ans b): _____

Case Study 4 (2 Marks)

Two sounds are produced with the same loudness but different shrillness.

Questions:

- a) Which physical quantity differs? (1)
- b) Name the unit of this quantity. (1)

Ans a) : _____

Ans b): _____

Case Study 5 (2 Marks)

A sound wave travels through air.

Questions:

- a) Name the type of wave. (1)
- b) Identify the regions of high and low pressure. (1)

Ans a) : _____



Ans b): _____

Case Study 6 (2 Marks)

A student observes that increasing amplitude affects how a sound is heard.

Questions:

- a) Which characteristic of sound changes? (1)
- b) What is its SI unit? (1)

Ans a) : _____

Ans b): _____

Case Study 7 (2 Marks)

Two tuning forks A and B have frequencies 256 Hz and 512 Hz respectively.

Questions:

- a) Which tuning fork produces shriller sound? (1)
- b) Give reason. (1)

Ans a) : _____

Ans b): _____

Case Study 8 (2 Marks)

An echo is heard after 0.8 s when a person shouts near a cliff.



Questions:

- a) What minimum distance is the person from the cliff? (1)
- b) State the condition necessary for echo formation. (1)

Ans a) : _____

Ans b): _____

Case Study 9 (2 Marks)

A sound wave has wavelength 0.85 m and frequency 400 Hz.

Questions:

- a) Calculate the speed of sound. (1)
- b) Identify the medium if possible. (1)

Ans a) : _____

Ans b): _____

Case Study 10 (2 Marks)

A sound travels faster in iron than in air.

Questions:

- a) Name the property of medium affecting speed of sound. (1)
- b) State one reason for this difference. (1)

Ans a) : _____

Ans b): _____



Case Study 11 (3 Marks – Tough)

A sound wave has frequency 500 Hz and speed 340 m s^{-1} .

Questions:

- a) Calculate its wavelength. (1)
- b) Identify whether the sound is audible, infrasonic, or ultrasonic. (1)
- c) State one use of such sound. (1)

Ans a) : _____

Ans b): _____

Ans c) : _____

Case Study 12 (3 Marks – Tough)

A person hears a thunder 5 s after seeing lightning.

Questions:

- a) Why is lightning seen earlier? (1)
- b) Calculate distance of the thunder source. (1)
- c) Name the type of wave involved. (1)

Ans a) : _____

Ans b): _____

Ans c) : _____

Case Study 13 (3 Marks – Tough)

Two sound waves X and Y have equal amplitudes but frequencies 300 Hz and 600 Hz.



Questions:

- Which wave is shriller? (1)
- Which wave has more energy? (1)
- Justify your answer. (1)

Ans a) : _____

Ans b): _____

Ans c) : _____

Case Study 14 (3 Marks – HOTS)

A student beats a drum lightly and then strongly.

Questions:

- Which property of sound changes? (1)
- What happens to amplitude? (1)
- Does pitch change? Give reason. (1)

Ans a) : _____

Ans b): _____

Ans c) : _____

Case Study 15 (3 Marks – HOTS)

An ultrasound scanner is used in hospitals.

Questions:

- Define ultrasound. (1)
- State one medical application. (1)
- Why ultrasound is preferred over X-rays in pregnancy? (1)



Ans a) : _____

Ans b): _____

Ans c) : _____

Case Study 16 (3 Marks – HOTS)

A ship uses SONAR to detect an underwater object.

Questions:

- Expand SONAR. (1)
- Write the formula used to calculate depth. (1)
- Name the type of waves used. (1)

Ans a) : _____

Ans b): _____

Ans c) : _____

Case Study 17 (3 Marks – HOTS)

A sound wave passes from air into water.

Questions:

- What happens to speed? (1)
- What happens to frequency? (1)
- What happens to wavelength? (1)

Ans a) : _____



Ans b): _____

Ans c) : _____

Case Study 18 (3 Marks – Conceptual)

Noise pollution is increasing in cities.

Questions:

- Define noise. (1)
- State one harmful effect. (1)
- Suggest one control measure. (1)

Ans a) : _____

Ans b): _____

Ans c) : _____

Case Study 19 (3 Marks – Numerical)

A sound source produces 680 vibrations in 2 s.

Questions:

- Calculate frequency. (1)
- Identify if sound is audible. (1)
- State audible range of humans. (1)

Ans a) : _____

Ans a) : _____



Ans b): _____

Case Study 20 (3 Marks – Competitive Level)

A sound wave travels 1700 m in 5 s.

Questions:

- Calculate speed of sound. (1)
- Identify the likely medium. (1)
- Give reason for your answer. (1)

Ans a) : _____

Ans b): _____

Ans c) : _____

End of Question Paper



Scans for solution

